

The equations of the tangent and the normal

Two lines at one point of a curve, from one derivative and one gradient relation.

LESSON 17–18

2 × 40 MIN

ALGEBRA 11, §1.6

P1 · 7.6

LESSON CLOCK

10'

20'

20'

22'

16'

12'

The three numbers you need	10 min
The tangent	20 min
The normal	20 min
Working backwards	22 min
Practice	16 min
Homework	12 min

LEARNING OBJECTIVES

- Write the equation of the tangent at a given point of a curve.
- Write the equation of the normal at that point.
- Find the point where a tangent has a given gradient.
- Find where a tangent meets the axes or the curve again.

TEXTBOOK

National	pp. 81–90
Cambridge	Pure Mathematics 1, pp. 159–167
Workbook	P1 Exercise 7F

01

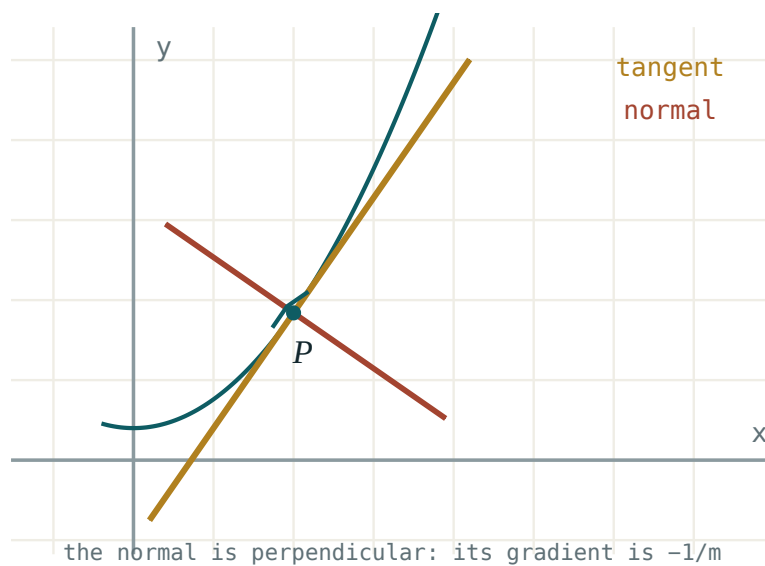
Explanation

Three numbers, then one formula

EVERY TANGENT QUESTION NEEDS THE SAME THREE

x_0 — where $y_0 = f(x_0)$ — the point · $m = f'(x_0)$ — the gradient. Then the point-gradient form finishes it:

$$y - y_0 = m(x - x_0)$$



The tangent touches; the normal crosses at a right angle. Both pass through the same point.

DIFFERENTIATE BEFORE SUBSTITUTING

Substituting x_0 into f first leaves a constant, whose derivative is zero. Find $f'(x)$ in full, and only then put the number in.

The tangent

Example. Find the tangent to $y = x^2 - 3x$ at $x = 2$.

the point	$y = 4 - 6 = -2$, so $(2, -2)$
the gradient	$y' = 2x - 3$, so $m = 1$
the line	$y + 2 = 1(x - 2)$
simplified	$y = x - 4$

A **horizontal** tangent occurs where $f'(x) = 0$; a **vertical** one where f' fails to exist by becoming unbounded.

The normal

The **normal** is the line through the same point, perpendicular to the tangent:

$$m_{\text{tangent}} \cdot m_{\text{normal}} = -1 \Rightarrow m_{\text{normal}} = -\frac{1}{f'(x_0)}$$

For the same curve at $(2, -2)$: the tangent gradient is 1 , so the normal gradient is -1 and the normal is $y = -x$.

TWO SPECIAL CASES

If $f'(x_0) = 0$ the tangent is horizontal and the normal is **vertical**: $x = x_0$, which has no gradient at all.

Writing $-\frac{1}{0}$ is not an answer.

Working backwards

Most examination questions give the gradient and ask for the point:

Example. Where does $y = x^3 - 4x$ have a tangent parallel to $y = 8x + 1$?

$$y' = 3x^2 - 4 = 8 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$$

Two points: $(2, 0)$ and $(-2, 0)$. Two tangents: $y = 8x - 16$ and $y = 8x + 16$.

PARALLEL AND PERPENDICULAR

Parallel to $y = mx + c$ means $f'(x) = m$. Perpendicular to it means $f'(x) = -\frac{1}{m}$. Both reduce to solving one equation.

02 Worked examples

EXAMPLE 1 Find the tangent and the normal to $y = x^3 - 2x$ at $x = 1$.

① $y(1) = -1$
Point $(1, -1)$.

② $y' = 3x^2 - 2, y'(1) = 1$

③ Tangent: $y + 1 = 1(x - 1)$.
 $y = x - 2$

④ Normal: gradient -1 .
 $y = -x$

ANSWER

tangent $y = x - 2$, normal $y = -x$

EXAMPLE 2 Find the tangent to $y = \frac{4}{x}$ at $x = 2$, and where it meets the axes.

① $y(2) = 2$
Point $(2, 2)$.

② $y' = -\frac{4}{x^2}, y'(2) = -1$

③ Tangent $y = -x + 4$.

④ Axes: $(4, 0)$ and $(0, 4)$.

ANSWER

$y = 4 - x$; meets the axes at $(4, 0)$ and $(0, 4)$

EXAMPLE 3 The tangent to $y = x^2$ at $x = a$ passes through $(0, -4)$. Find a .

① Point (a, a^2) , gradient $2a$.

② $y - a^2 = 2a(x - a)$

③ At $(0, -4)$: $-4 - a^2 = -2a^2$.

④ $a^2 = 4 \Rightarrow a = \pm 2$

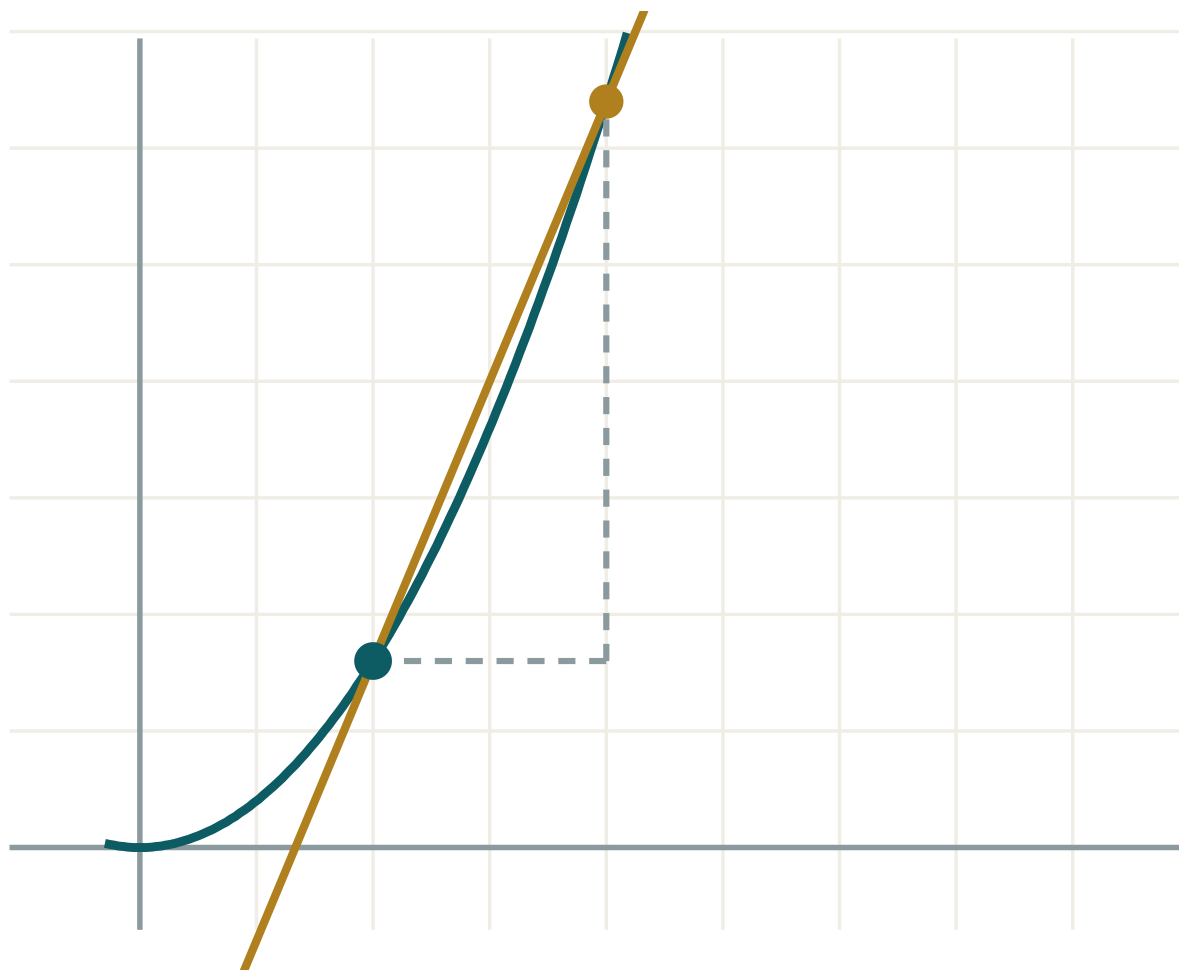
ANSWER

$$a = \pm 2$$

03 Interactive model

Draw both lines on one sketch — students who see only the tangent forget the normal is perpendicular.

Tangent at a point


 x 2

 h 2

 $\Delta y / \Delta x$ 2.4

 $f'(x)$ exactly 1.6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Tangent	Urinma	Касательная
Normal	Normal	Нормаль
Point of tangency	Urinish nuqtasi	Точка касания
Gradient of a tangent	Urinma burchak koeffitsienti	Угловой коэффициент касательной
Perpendicular gradients	Perpendikulyar koeffitsientlar	Взаимно перпендикулярные
Point-gradient form	Nuqta-koeffitsient shakli	Уравнение через точку и угловой коэффициент
Horizontal tangent	Gorizontal urinma	Горизонтальная касательная
Intercept	Kesim	Отрезок на оси

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The gradient of the tangent at x_0 is:

$$f(x_0)$$

$$f'(x_0)$$

$$-\frac{1}{f'(x_0)}$$

$$0$$

The normal gradient is:

$$f'(x_0)$$

$$-\frac{1}{f'(x_0)}$$

$$\frac{1}{f'(x_0)}$$

$$0$$

A horizontal tangent occurs where:

$$f' = 1$$

$$f' = 0$$

$$f = 0$$

$$f' \text{ is undefined}$$

If $f'(x_0) = 0$ the normal is:

horizontal

vertical

$y = 0$

undefined

Tangent to $y = x^2$ at $x = 3$:

$y = 6x - 9$

$y = 6x + 9$

$y = 3x$

$y = 9x - 6$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Tangent to $y = x^2$ at $x = 1$

ANSWER $y = 2x - 1$

2. Normal to $y = x^2$ at $x = 1$

ANSWER $y = -0.5x + 1.5$

3. Tangent to $y = 3x + 2$ at $x = 0$

ANSWER $y = 3x + 2$

4. Gradient of the tangent to $y = x^3$ at $x = 2$

ANSWER 12

5. Where is the tangent to $y = x^2 - 6x$ horizontal?

ANSWER $x = 3$

6. Tangent to $y = x^2$ at $x = 0$

ANSWER $y = 0$

7. Normal to $y = x^2$ at $x = 0$

ANSWER $x = 0$

Medium · the standard question

7 PROBLEMS

1. Tangent to $y = x^3 - 2x$ at $x = 1$

ANSWER $y = x - 2$

2. Normal to the same at the same point

ANSWER $y = -x$

3. Tangent to $y = \frac{4}{x}$ at $x = 2$

ANSWER $y = 4 - x$

4. Tangent to $y = \sqrt{x}$ at $x = 4$

ANSWER $y = \frac{x}{4} + 1$

5. Where has $y = x^3 - 4x$ a tangent parallel to $y = 8x$

ANSWER $x = \pm 2$

6. Where has $y = x^2 - 4x$ a tangent of gradient 2

ANSWER $x = 3$

7. Tangent to $y = x^2 - 5x + 6$ at its x -intercepts

ANSWER $y = -x + 2$ and $y = x - 3$

Hard · stretch and reason

7 PROBLEMS

1. Tangent to $y = x^2$ through $(0, -9)$

ANSWER $y = \pm 6x - 9$

2. Normal to $y = x^2$ at $x = 1$ meets the curve again at

ANSWER $x = -1.5$

3. Find k so $y = kx$ is tangent to $y = x^2 + 4$

ANSWER $k = \pm 4$

4. Tangent to $y = \frac{1}{x}$ at $x = a$; show the axes cut a triangle of constant area

ANSWER Area 2 always

5. Where do the tangents to $y = x^3$ at $x = 1$ and $x = -1$ meet?

ANSWER $(0, -2)$

6. Normal to $y = \sqrt{x}$ at $x = 1$: find its equation

ANSWER $y = -2x + 3$

7. Two curves $y = x^2$ and $y = \frac{1}{x}$ meet at $x = 1$. Find the angle between them

ANSWER $\approx 71.6^\circ$

07 Homework

Homework — 5 tasks

Give both lines in the form $y = mx + c$, simplified.

- Find the tangent and the normal to $y = x^2 - 4x + 1$ at $x = 3$.
- Find the tangent to $y = \frac{6}{x}$ at $x = 3$ and where it meets the axes.
- Find the points on $y = x^3 - 3x^2$ where the tangent is horizontal.
- Find where $y = x^3 + x$ has a tangent parallel to $y = 4x - 7$.

5. Find the tangent to $y = x^2$ that passes through $(0, -1)$.